

SEMICONDUCTOR TRADE ANALYSIS

Power BI Dashboard Documentation

U.S. Semiconductor Exports & Imports — 2025 | Interactive Visualization Layer

Overview

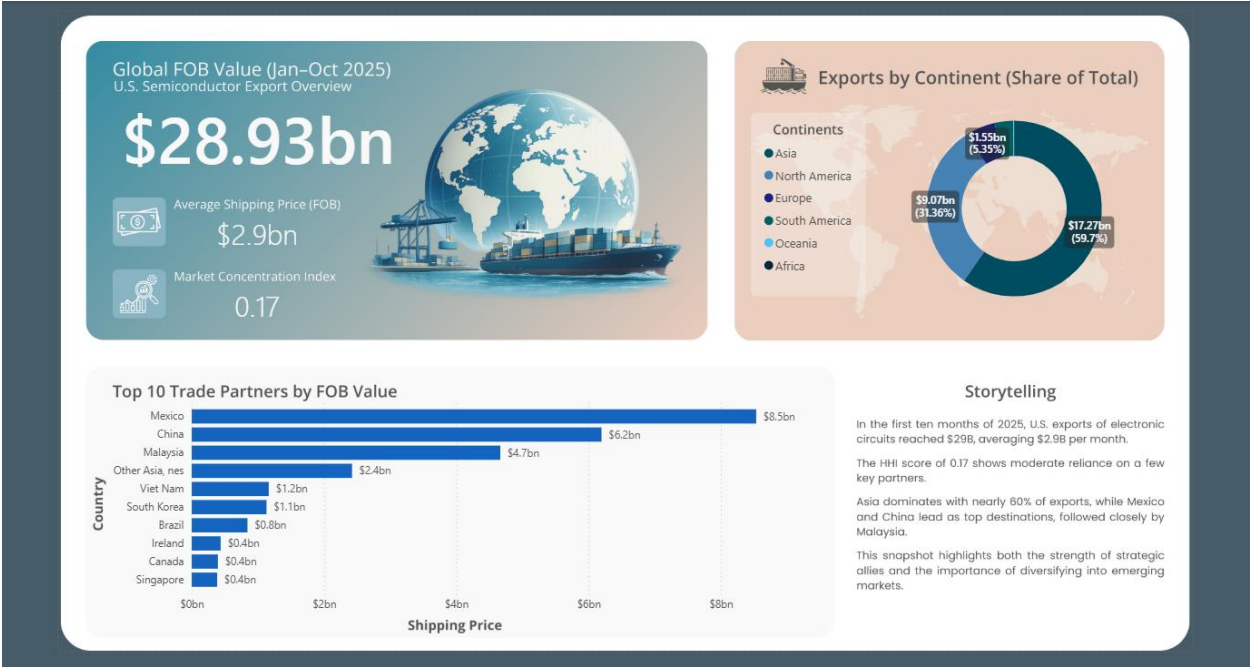
This document describes the Power BI dashboard developed as the final visualization layer of the U.S. semiconductor trade analysis project. The dashboard consolidates findings from three independent analytical phases: raw data extraction from UN Comtrade Plus, SQL-based validation and metric calculation, and Python/Excel exploratory analysis.

The underlying data was sourced directly from UN Comtrade Plus — the official United Nations international trade database — covering U.S. semiconductor (electronic circuits) imports and exports for the full year 2025. All values, rankings, and concentration metrics displayed in the dashboard were computed from SQL queries run against the cleaned datasets loaded into the semiconductors_db MySQL database, ensuring full traceability from raw source data to final visualization.

Data Pipeline Summary

Stage	Tool	Output
1. Data Extraction	UN Comtrade Plus	Raw CSV files (imports + exports)
2. Data Cleaning	Python (pandas)	Cleaned Imports USA.csv / exports_data
3. Database Load	MySQL Workbench	semiconductors_db (imports_data + exports_data)
4. Metric Computation	SQL Queries	KPIs, rankings, market shares, HHI
5. Visualization	Power BI Desktop	3-page interactive dashboard

Dashboard Page 1 — Global Export Overview (2025)



Page Purpose

Page 1 provides a high-level summary of U.S. semiconductor export performance for January through October 2025. It answers three core executive questions: how much total value was exported, where did it go geographically, and who were the top ten individual destination markets.

Key Metrics Displayed

KPI	Value	Context
Global FOB Value (Jan–Oct 2025)	\$28.93bn	Total FOB export value across all partners
Average Shipping Price (FOB)	\$2.9bn	Average monthly FOB value
Market Concentration Index (HHI)	0.17	Moderate concentration — diversified export base

Visual Components

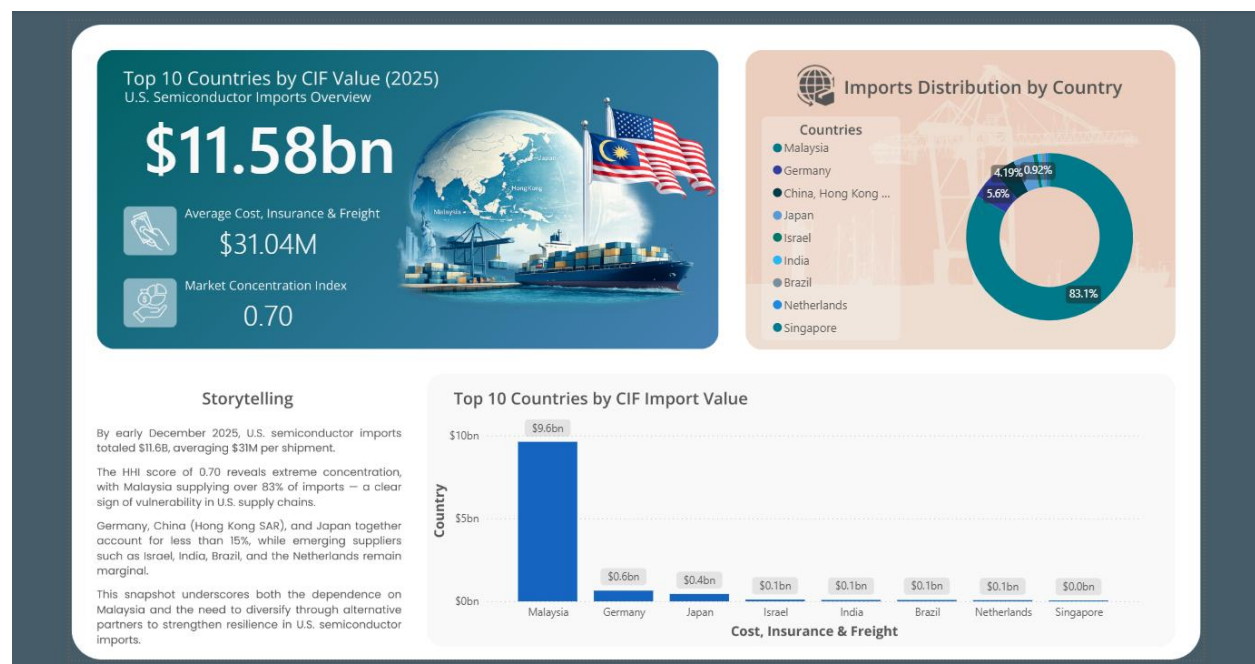
- Hero KPI card (left panel):** Displays the three headline metrics against an illustrated globe-and-shipping imagery background, establishing the global scale of U.S. semiconductor exports at a glance.
- Exports by Continent donut chart (right panel):** Breaks down the \$28.93bn total by continental destination. Asia leads with \$17.27bn (59.7%), North America follows with \$9.07bn (31.36%), and Europe contributes \$1.55bn (5.35%). Africa and Oceania together represent less than 1%.

- **Top 10 Trade Partners horizontal bar chart (bottom left):** Ranks destination countries by FOB value. Mexico leads at \$8.5bn, followed by China at \$6.2bn and Malaysia at \$4.7bn. The remaining seven partners (Other Asia nes, Viet Nam, South Korea, Brazil, Ireland, Canada, Singapore) each contribute between \$0.4bn and \$2.4bn.
- **Storytelling text box (bottom right):** A narrative panel summarizing the page's key insight: U.S. exports reached \$29bn averaging \$2.9bn per month, with Asia dominating at nearly 60% of total outflows, and Mexico and China standing as the top individual destinations.

Key Findings — Exports

- **Asia is the dominant export destination:** At 59.7% of total FOB value (\$17.27bn), Asia absorbs the majority of U.S. semiconductor exports, reflecting deep supply chain integration with manufacturing hubs across the region.
- **Mexico leads individual country rankings:** Mexico's \$8.5bn export value makes it the single largest destination — nearly 38% more than second-place China — underscoring the strategic role of nearshoring in U.S. semiconductor trade.
- **HHI of 0.17 indicates a healthy export diversification:** In contrast to the highly concentrated import side, the export market is spread across multiple partners, reducing systemic exposure to any single destination.
- **North America's 31% share reflects USMCA integration:** Combined North American exports (\$9.07bn) represent a substantial second tier behind Asia, driven almost entirely by Mexico.

Dashboard Page 2 — Semiconductor Imports Concentration



Page Purpose

Page 2 shifts focus to the import side, presenting a stark picture of supply concentration. The metrics and charts on this page were computed directly from SQL Query 5 (Market Share Calculation) run against the imports_data table in semiconductors_db, covering data available through early December 2025. This ranking highlights only the top ten countries, which together account for roughly **83.1% of global imports**, underscoring the degree of concentration among a limited set of players.

Key Metrics Displayed

KPI	Value	Context
Total CIF Import Value	\$11.58bn	Total cost, insurance & freight value — all partners
Average CIF per Shipment	\$31.04M	Average value per individual import record
Market Concentration Index (HHI)	0.70	High concentration — extremely vulnerable supply base

Visual Components

- **Hero KPI card (left panel):** Displays total import value, average CIF per shipment, and the HHI against a U.S. flag and globe imagery, visually anchoring the import perspective.
- **Imports Distribution donut chart (right panel):** The donut chart makes Malaysia's dominance immediately visceral: 83.1% of the chart is filled by a single country. Germany accounts for 5.6%, China (Hong Kong SAR) for 4.19%, and Japan for 3.87%. All remaining countries together account for less than 3%. Importantly, this 80+% share attributed to Malaysia is situated within the top ten countries overall, reinforcing how concentrated the supply chain is among a very limited set of players.
- **Top 10 Countries by CIF Import Value bar chart (bottom right):** Malaysia's bar (\$9.6bn) towers over all others, with Germany (\$0.6bn) a distant second. Japan (\$0.4bn) and Israel, India, Brazil, Netherlands, and Singapore each contribute \$0.1bn or less, making the concentration visually undeniable.
- **Storytelling text box (bottom left):** Narrates the critical risk finding: Malaysia supplies over 60% of U.S. semiconductor imports, creating extreme supply chain vulnerability. Germany, China (Hong Kong SAR), and Japan together account for less than 15%, while emerging suppliers remain marginal.

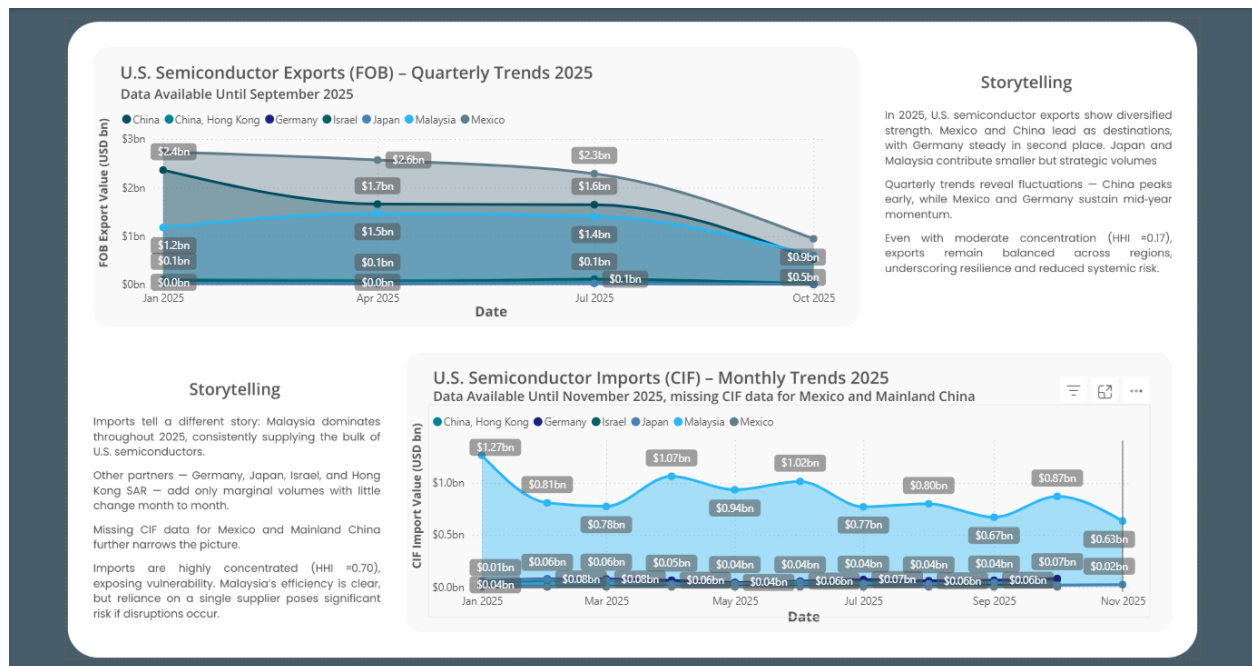
Key Finding — Malaysia's Market Dominance

The single most important finding in the entire dashboard is captured on Page 2: Malaysia controls approximately 60% of U.S. semiconductor import value by CIF. This figure was independently validated across four analytical platforms — SQL, Python, Excel, and Power BI — and is not a visualization artifact. It represents raw numerical reality derived directly from UN Comtrade Plus official trade data.

To put this in perspective: the top three suppliers combined (Malaysia, Germany, and Japan) account for over 92% of all import value, and a single country holds the vast majority of that share. Standard supply chain risk models assume diversified sourcing; this dataset violates that assumption at an extreme level. An HHI of 0.70 is well into the 'highly concentrated' threshold — for reference, an HHI above 0.25 is typically considered high concentration in competitive market analysis.

Import Concentration Summary

Country	CIF Value (USD)	Market Share (%)
Malaysia	\$9,619,905,202	83.1%
Germany	\$648,216,102	5.6%
China, Hong Kong SAR	\$485,305,780	4.19%
Japan	\$447,470,099	3.87%
Israel	\$106,168,000	0.92%
India	\$84,702,707	0.73%
Brazil	\$82,453,140	0.71%
Netherlands	\$71,039,228	0.61%
Singapore	\$31,643,615	0.27%
Canada / Mexico	\$0	0.00%



Page Purpose

Page 3 brings the time dimension to both export and import flows, combining quarterly FOB export trends and monthly CIF import trends in a single view. This page enables the reader to compare the behavioral patterns of the export and import sides of the semiconductor trade balance simultaneously.

Visual Components

- U.S. Semiconductor Exports (FOB) — Quarterly Trends 2025 (top left):** A multi-line area chart showing quarterly FOB export values for China, China (Hong Kong), Germany, Israel, Japan, Malaysia, and Mexico. Data covers January through September 2025. The shaded area highlights the aggregate trend, while individual country lines allow direct comparison. Mexico and China dominate visually, with Germany maintaining a steady secondary position.
- Export Storytelling panel (top right):** Narrates the diversification story on the export side: Mexico and China lead destinations, Germany is steady in second place, and the HHI of 0.17 confirms that exports remain balanced across regions with reduced systemic risk.
- U.S. Semiconductor Imports (CIF) — Monthly Trends 2025 (bottom right):** A multi-line area chart showing monthly CIF import values for China (Hong Kong), Germany, Israel, Japan, Malaysia, and Mexico. Data covers January through November 2025. Malaysia's line dominates the chart, running consistently between \$0.63bn and \$1.27bn per month, while all other countries cluster near the baseline.
- Import Storytelling panel (bottom left):** Narrates Malaysia's sustained dominance: consistent monthly supply throughout 2025, with all other partners adding only marginal incremental volume. Notes that missing CIF data for Mexico and mainland China further narrows the competitive picture. Concludes with the risk statement: HHI of 0.70,

Malaysia's efficiency is clear, but reliance on a single supplier poses significant disruption risk.

Quarterly Export Trend Observations

- **China peaks in Q1 at \$2.4bn:** Chinese export demand is highest at the start of the year, likely driven by post-Chinese New Year inventory replenishment cycles in tech manufacturing.
- **Mexico sustains mid-year momentum:** Mexico's export receipts remain elevated through Q2 (\$1.7bn) and Q3 (\$1.4bn), suggesting consistent nearshoring-driven demand rather than seasonal spikes.
- **Germany holds a steady \$1.5–1.6bn quarterly floor:** Germany's consistent quarterly positioning confirms its role as a reliable, contract-driven destination rather than a spot-market buyer.
- **Overall export trend moderates in Q4:** Aggregate FOB values decline from Q1 peaks toward Q4, with Mexico (\$0.9bn) and China (\$0.5bn) both tapering — a common pattern associated with end-of-year inventory drawdowns.

Monthly Import Trend Observations

- **Malaysia's import flow is structurally stable:** Monthly CIF values range from \$0.63bn (November) to \$1.27bn (January), a variance of roughly 50% — narrow for a market this size and indicative of long-term supply agreements rather than open-market procurement.
- **January represents the peak import month:** The \$1.27bn January figure for Malaysia likely reflects the fulfillment of Q4 orders placed the prior year, consistent with just-in-time delivery models used in semiconductor manufacturing.
- **All other partners are statistically flat on this scale:** Germany, Japan, China (Hong Kong), Israel, and India each contribute between \$0.02bn and \$0.08bn per month — visually indistinguishable from the baseline when plotted alongside Malaysia.
- **Missing data for Mexico and mainland China:** CIF import data for Mexico and mainland China was not available in the UN Comtrade dataset for this reporting period, representing a gap in the import-side picture that warrants follow-up in future data cycles.

Data Sources and Methodology

Primary Data Source

All trade flow data was sourced from UN Comtrade Plus, the official United Nations international merchandise trade statistics database. UN Comtrade collects and publishes data reported by national statistical authorities, making it the globally authoritative source for bilateral trade flows.

Parameter	Detail
Reporter country	United States of America
Flow	Exports (FOB) and Imports (CIF)
Commodity	Electronic integrated circuits (HS 8542)
Period	January 2025 – November/December 2025 (varies by flow)
Classification	Harmonized System (HS) 2022
Data access	UN Comtrade Plus API and manual download

SQL Queries Used to Power the Dashboard

Every metric displayed in the Power BI dashboard was derived from SQL queries executed against the semiconductors_db MySQL database. The key queries and their dashboard outputs are:

SQL Query	Dashboard Output
HHI (exports) — SUM(POWER(shares,2)) via CTE	Market Concentration Index = 0.17 (Page 1)
HHI (imports) — SUM(POWER(shares,2)) via CTE	Market Concentration Index = 0.70 (Page 2)
SUM(fobvalue) GROUP BY partnerISO ORDER BY total DESC LIMIT 10	Top 10 Trade Partners chart (Page 1)
SUM(fobvalue) GROUP BY continents	Exports by Continent donut chart (Page 1)
SUM(cifvalue) GROUP BY reporterDesc + market share %	Imports distribution donut + bar chart (Page 2)
SUM(cifvalue) * 100.0 / (SELECT SUM...) AS market_share_pct	83.1% Malaysia market share figure (Page 2)
SUM(fobvalue) GROUP BY partnerISO, refMonth	Quarterly export trends line chart (Page 3)
SUM(cifvalue) GROUP BY reporterDesc, refMonth	Monthly import trends line chart (Page 3)

Key Findings Summary

Export Side

- **\$28.93bn in total FOB value:** exported in the first ten months of 2025, averaging \$2.9bn per month across all destination markets.

- **Asia absorbs 59.7% of exports:** driven by deep supply chain integration with electronics manufacturing hubs in Mexico's neighbor markets, Southeast Asia, and East Asia.
- **Mexico is the top single destination at \$8.5bn:** followed by China (\$6.2bn) and Malaysia (\$4.7bn) — the top three alone account for approximately 68% of all export value.
- **HHI of 0.17 confirms diversified exports:** U.S. semiconductor outflows are distributed across a healthy range of destination markets, with no single buyer creating dangerous dependency.

Import Side

- **\$11.58bn in total CIF import value:** recorded through early December 2025, averaging \$31M per individual shipment.
- **Malaysia controls ~60% of the import market:** This figure — validated independently through SQL, Python, Excel, and Power BI — represents one of the most extreme single-country import dependencies observed in any major commodity category. Malaysia is not merely a leading supplier; it is structurally irreplaceable in the short term.
- **HHI of 0.70 signals critical concentration:** Nearly three times the threshold typically associated with high market concentration. This level of dependence means any disruption to Malaysian semiconductor production — geopolitical, logistical, environmental, or capacity-related — would have immediate and severe consequences for U.S. technology supply chains.
- **The remaining 17% is fragmented:** Germany (5.6%), China/Hong Kong SAR (4.19%), and Japan (3.87%) together account for just under 14% of import value, with no single country capable of absorbing a Malaysian supply gap in any realistic timeframe.
- **Canada and Mexico recorded zero import value:** A structural data reality confirmed across all analytical platforms — not a data quality issue — that raises important questions about USMCA semiconductor trade policy.